

Backend Service & Queue Optimization Engineer

Focus: Dynamic Queue Wait-Time Algorithm, Mandi Load Balancing & Data Engine

Executive Summary & Core Responsibility:

Engineered mathematical wait-time heuristics, multi-mandi load balancing routing, and notification dispatch services.

1. Technology Stack & Architectural Components

Domain / Layer	Specific Technologies, Libraries & Frameworks Used
Algorithmic Engine	Python Heuristics, Mathematical Modeling (M/M/c queueing approximation)
Data Layer	SQLite3 relational schemas, automated seed data generation, transactional logging
Validation & Modeling	Pydantic v2 (Strict Request Schemas & Dynamic Typing)
Notification Services	Simulated Multi-Channel Messaging (SMS, WhatsApp, In-App Audit Trail)
Business Logic	Official 2026 MSP rate indexing, Gross-to-Tare weight differential mathematics

2. Key Modules Developed & Production Deliverables

- **Dynamic Wait-Time Estimation Algorithm (queue_service.py):** Developed predictive calculation taking into account vehicle type factors (Tractors: 1.0x, Large Trucks: 1.8x, Pickups: 0.7x), crop lab durations, and active weighbridge lanes.
- **Mandi Load Balancer & Congestion Routing (centers.py):** Formulated intelligent routing recommending alternate nearby mandis when primary center wait time exceeds 45 minutes.
- **Database Architecture & Seeding (database.py):** Created optimized schema with tables for centers, farmers, procurement tokens, stage logs, and notification archives with 7+ realistic live demo records.
- **Multi-Channel Notification Dispatcher (notification_service.py):** Built automated SMS and stage milestone notification engine triggering alerts at every progression point.

3. Direct Codebase Ownership & Modified Files

Source Files:

backend/app/services/queue_service.py • backend/app/services/notification_service.py • backend/app/database.py • backend/app/routers/centers.py • backend/app/routers/farmers.py

4. SIH Jury Defense & Live Evaluation Points (What to Speak Tomorrow)

- **Presentation Pitch Point:** Walk through the formula: $T_{service} = \max(T_{weigh} / N_{scales}, T_{quality} / N_{labs}) * Vehicle_Factor$.
- **Presentation Pitch Point:** Explain how the database maintains complete auditability with immutable stage_logs for every kg weighed.
- **Presentation Pitch Point:** Highlight how the load balancing engine prevents bumper-to-bumper traffic jams outside APMC mandis.